



[BIT703 Web Technologies](#) / [Contents](#) / Topic 7: Client-side web application development and programming – JavaScript / UI frameworks for client-side development

How to pick an appropriate framework

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We have introduced three of the most prevalent JavaScript frameworks used in industry, but they are not the only ones available. There are several frameworks or just libraries that you can choose from. It is important to choose the right framework for the project based on the web application requirements and not just because everyone is using a particular framework. Things to consider are:

► Documentation

A good framework should include clear and detailed documentation to help you use its features effectively. If the documentation isn't sufficient, you'll need access to other resources like blogs, forums or video tutorials. Poor documentation and limited support often cause frameworks to lose popularity.

► Support

Support is just as important as documentation. It includes regular updates, bug fixes, new features and compatibility with other code, such as Node.js modules. You want a framework that will remain reliable and relevant for several years. It can be frustrating if a framework meets all your requirements but loses support shortly after you start using it.

► Ease of use and integration

Even with good documentation, a framework isn't helpful if it's hard to integrate into your application. For example, if you don't plan to use Node.js, React wouldn't be a suitable library because it's difficult to use without Node.js. This is owing to the requirement of the npm (node package manager) to set the React environment (refer to the React installation documentation).

► Features

While features aren't the most critical factor, they still matter. You should choose a

[Skip to main content](#)

framework that offers the features you need, such as usability, performance, dependencies and compatibility with other libraries.

LambdaTest considers these the 27 Best JavaScript Frameworks for 2025 (Gadhiya, 2025)^{fnref:39afad}. These frameworks fall into three main categories:

- **Frontend frameworks:** Most frameworks belong to this group. They focus on the user interface but often rely on another language, like PHP, to handle backend or server-side operations.
- **Backend frameworks:** These frameworks manage backend or server-side operations, replacing the need for languages like PHP.
- **Testing frameworks:** These frameworks focus on testing web applications using automated testing methods.

Choosing the right framework requires careful consideration of your project's needs and long-term goals.



☒ Framework selection challenge

Non-Assessed

[Skip to main content](#)

Imagine you are part of a development team tasked with building a dynamic, interactive e-commerce website. The site needs to have the following capabilities:

1. A highly responsive and visually appealing user interface for browsing products
2. Real-time updates to reflect stock availability, discounts and personalised recommendations
3. A backend that handles secure transactions and user authentication
4. Automated testing to ensure smooth functionality during every update.

Your goal is to select the most suitable frameworks (frontend, backend and testing) from the options below:

Available frameworks

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- **React:** A frontend framework renowned for creating highly interactive user interfaces. Best paired with JavaScript/Node.js.
 - **AngularJS:** A frontend framework known for building full-featured and large-scale web applications. Comes with built-in tools for testing.
 - **Express.js:** A minimal and flexible Node.js framework for backend development, offering performance and scalability.
 - **Django:** A Python-based backend framework with strong support for secure transactions and authentication features.
 - **Mocha:** A JavaScript-based testing framework, ideal for automated testing of frontend and backend code.
 - **Selenium:** A testing framework popular for end-to-end testing across web applications.

Your task:

1. Review the requirements for the e-commerce website and the available frameworks.
2. Choose **one front-end framework, one back-end framework** and **one testing framework** that best suit the scenario.

Skip to main content

3. Justify your selections briefly (1–2 sentences for each framework).

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Thoughts on this page?



Discussion

With so many JavaScript frameworks in the mix, how do developers crack the code to find *the one* that perfectly aligns with their project's needs, wows on features and promises to stand the test of time? Share your views with your peers.

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Jorge Andres Guerrero Giraldo 14 Jan 2026, 2:43 PM

Developers choose a JavaScript framework by matching it to the project's requirements, team skills, and long term maintenance needs. I start by defining what matters most (performance, SEO, routing, state complexity, accessibility, and expected scale), then compare frameworks using a small proof of concept to validate developer experience and fit. I also look at "future proofing" signals such as community size, update frequency, documentation quality, ecosystem maturity, and the availability of testing, build, and security tooling. Finally, I prefer frameworks that reduce lock in, follow web standards, and have a clear upgrade path, because long term success is usually about maintainability, not just features.

